

3Dify: Extruding Common 2D Charts with Timeseries Data

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ABSTRACT

3D charts are not common in financial services. We review chart use in practice. We create 3D financial visualizations starting with 2D charts used extensively in financial services, then extend into the third dimension with timeseries data. We embed the 2D view into the 3D scene; constrain interaction and add depth cues to facilitate comprehension. Usage and extensions indicate success.

Index Terms: Human-centered computing—Visualization—Visualization application domains—Information visualization Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

A global financial firm wanted engaging insightful visualization of financial data on large scale displays (meeting room displays, multi-touch walls, and a CAVE). Prior visualizations were unsuccessful due to confusing 3D, missing cues such as tick labels and grid lines; or very plain 2D lacking visceral appeal.

The target 3D visualization requirements included: a) ease of understanding (i.e. immediate comprehension, no tutorials, no complex legends); b) ease of interaction; c) no extra devices (e.g. no 3D glasses, VR, mobile AR, etc). To create a successful application, our method reviewed many design approaches (section 2); created an extensible design framework (3), and tracked user acceptance (4). Our contribution is a framework for interactive 2D point-in-time charts extruded with timeseries data into the third dimension.

2 BACKGROUND

Financial visualizations including 2D, 3D, VR and AR [4, 8, 9] can be organized as follows (see poster for more citations):

3D surfaces plot financial instruments modeled as multidimensional functions (futures, options, etc.), e.g. VR worlds within worlds [5], forex [6], and financial software, e.g. Bloomberg.

Aligned multi-component 3D visualizations visually align components along shared axes, e.g. price, time, categories [10].

3D spatialized datasets plot a singular dataset into a Cartesian space, e.g. scatterplots, tables, glyphs, bars, timeseries, ribbons, or generalized atomic elements. Data may be indirectly mapped to spatial coordinates, e.g. treemaps, graphs, trees or spheres.

AR & physical spaces: NYSE has a 3D trading floor [3]. WSJ's has an AR stock market anchored in a human-sized space [7].

3D workspaces of 2D screens & charts place 2D screens (watchlists, news, spreadsheets, tv) in a 3D virtual space, e.g. [2].

Financial professionals' charts differ from academic researchers' financial visualization. We reviewed 25 financial publications with more than 1300 charts. Table 1 summarizes the top chart types, covering 99.5% of the charts. There were no 3D charts. Timeseries charts are most prevalent, most other charts are "point-in-time."

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Table 1: Popular Charts in 25 Financial Publications.

Chart type	Num charts	Pct charts	Num pubs
Timeseries (line/bar)	958	71.5	25
Stack/cluster bar	116	8.7	20
Simple bar	107	8.0	14
Pie/donut	27	2.0	11
Scatterplot/bubble	20	1.5	11
Boxplot / range	33	2.5	8
Curve (e.g. yield)	17	1.3	6
Histogram	5	0.4	5
Heatmap/rank grid	45	3.4	4

3 ANALYSIS, DESIGN AND DEVELOPMENT

Given the mismatch between researchers' visualizations and financial analysts', we focused on the analysts' 2D charts as the starting point. With the exception of timeseries charts, most other charts show data at a point in time. Given the prevalence of timeseries charts (71% of all charts), we considered how to add temporal data into familiar pre-existing 2D charts. *Animation* was dismissed as it shows time incrementally making it difficult to see temporal patterns. *Data comics* were dismissed as many panels would clutter the display, reduce font sizes, and require more reading effort. *3D* adds temporal data in the third dimension, but has issues [1], such as:

Poor 2D views result from 3D vis designed without regard to 2D viewpoints. 2D views have overlapping labels, overplotted data, etc.

Disorientation results from freeform navigation. The scene may be backwards, or break conventions, e.g., time plotted left-to-right.

Perspective cues may be missing (e.g. grids); or ambiguous (e.g., two lines of the same pixel color at different depths.)

Text ambiguity results when text overlaps, or not differentiated from background, etc. Labels may be difficult to associate with axes.

2D charts extruded with time data. We start with the canonic 2D chart view, then add 3D, only when rotated (Fig. 1). Only one additional dimension, time, is added, making it easy to conceptualize the addition without significantly increasing cognitive load. The 2D chart, when rotated, retains the expected labels, ticks, borders and text oriented in the plane of the 2D chart.

Camera constraints & 2D snap. Camera motion is constrained to an octant, shown by the yellow area in Fig. 2A. The viewer can rotate the scene within the octant, accessing a 2D front view, 2D side view and 2D top view; and any 3D viewpoint in between. The front 2D view is the canonic point-in-time chart, while the side 2D view is a canonic financial timeseries chart. When the camera aligns with a 2D view, all the non-relevant 3D chart content (data, grids, ticks, axes labels), is removed; creating a perfectly flat, canonic 2D chart with no extraneous items in the scene.

3D cues. Binocular depth cues nor interactions such as rotation to trigger motion parallax could not be relied on. Grids on floors and walls help locate items; and fog differentiates foreground and background, e.g., lines crossing themselves in the wormplot appear more saturated in the foreground (Fig. 2C). Surfaces, such as ribbons on the extruded bar chart, or a loft between two curves in the wormplot, allows for the use of lighting cues, such as change in brightness and specular highlights to help indicate the orientation of a patch.

Animation & interaction. In ambient mode, the system uses a bouncing-ball animation to trigger interactions to engage viewers.

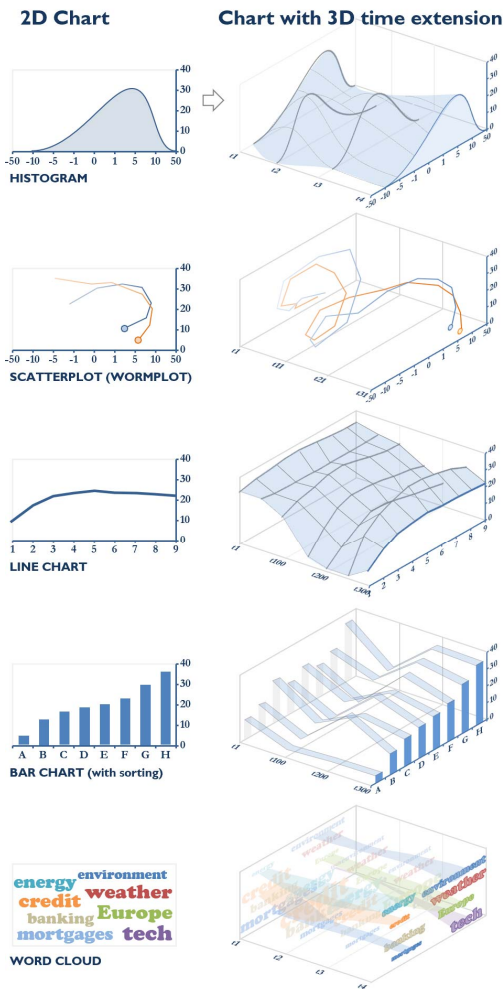


Figure 1: Five common 2D visualizations, and their 3D counterparts with added timeseries data.

Viewers can rotate view points, change datasets, select series, change date range, etc. Labels will re-orient to different planes as the rotation approaches the side view or top view to remain legible.

Extensions to VR, AR, Mobile and 3D Printing. The system was designed for interactive large scale ambient displays and extended for a CAVE, tablet and 3D printing (Fig. 2B).

4 DISCUSSION

A bouncing-ball style overlay indicated available interactions such as menu taps to change data, selection, 3D rotation and other interactions. Event logging indicated 3D rotation the top interaction at nearly 40%. Anecdotal evidence from project managers indicated positive response to the use of 3D among target audiences, including staff, executive guests and educational visitors. Sponsors validated the project through scope and funding: the initial plan was for two 2D/3D visualizers; which expanded to 5 visualizers plus VR, mobile and 3D printing.

5 CONCLUSION AND FUTURE WORK

Although 3D has been in finance for 30 years, we believe widespread adoption of 3D in the domain will need to leverage incremental approaches to bridge between 2D and 3D. Future work should evaluate

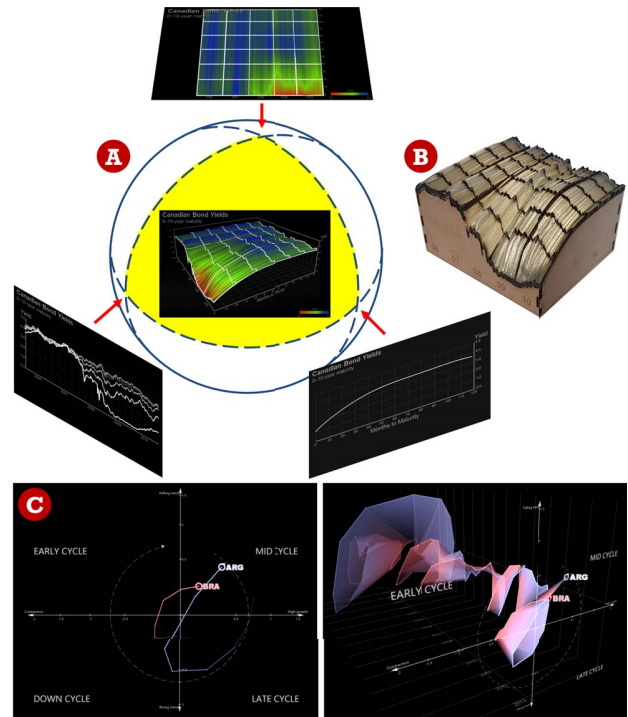


Figure 2: A. 2D charts occur at the limits of the viewer constraints and a 3D view occurs in-between. B. 3D print of same dataset. C. Wormplot in 2D and 3D, with longer timeseries and depth cues

how embedded 2D canonic charts aid user interpretation of 3D. Further, some 2D chart types will not effectively extrude with time (pie, stacked bar, radar, etc), e.g. due to occlusion.

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REFERENCES

- [1] R. Brath. 3d infovis is here to stay: Deal with it. In *2014 IEEE VIS International Workshop on 3DVis (3DVis)*, pp. 25–31. IEEE, 2014.
- [2] E. Carson. Quantvr wants to turn stock market data into immersive virtual reality experiences, 2015.
- [3] B. Delaney. The nyse’s 3d trading floor. *IEEE Computer Graphics and Applications*, 19(6):12–15, 1999.
- [4] M. Dumas, M. J. McGuffin, and V. L. Lemieux. Financevis. net-a visual survey of financial data visualizations. In *Poster Abstracts of IEEE Conference on Visualization*, vol. 2, p. 8, 2014.
- [5] S. K. Feiner and C. Beshers. Worlds within worlds: Metaphors for exploring n-dimensional virtual worlds. In *Proceedings of the 3rd annual ACM SIGGRAPH symposium on User interface software and technology*, pp. 76–83, 1990.
- [6] D. L. Gresh, B. E. Rogowitz, M. Tignor, and E. Mayland. An interactive framework for visualizing foreign currency exchange options. In *Proceedings Visualization’99 (Cat. No. 99CB37067)*, pp. 453–562. IEEE, 1999.
- [7] R. Kenny. Introducing the wall street journal stock market on magic leap, 2019.
- [8] A. Lugmayr, Y. J. Lim, J. Hollick, J. Khuu, and F. Chan. Financial data visualization in 3d on immersive virtual reality displays. In *International Workshop on Enterprise Applications, Markets and Services in the Finance Industry*, pp. 118–130. Springer, 2018.
- [9] Z. Niu, R. Li, J. Wu, Y. Xue, and J. Zhang. regvis. net—a visual bibliography of regulatory visualization. *arXiv:2007.03573*, 2020.
- [10] W. Wright. Information animation applications in the capital markets. In *Proceedings of Visualization 1995 Conference*, pp. 19–25. IEEE, 1995.